

Replication Plan

Common-Mode Voltage Reduction of Single-Phase Quasi-Z-Source Inverter-Based
Photovoltaic System

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Auto-generated Replication Blueprint

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Paper Overview

This paper proposes and analyzes modulation strategies for reducing common-mode voltage (CMV) in a single-phase quasi-Z-source inverter (qZSI) used in transformerless photovoltaic (PV) grid-tied systems. The high CMV in conventional modulation causes undesirable ground leakage current through the PV panel parasitic capacitance. The authors derive the CMV expressions for different switching states, propose modified PWM schemes that avoid high-CMV states, and validate through circuit simulation showing reduced CMV and leakage current while maintaining output power quality.

Detailed Workflow

Phase 1: Circuit Topology Modeling

1.1 Model the quasi-Z-source inverter.

- Define the qZSI topology: two inductors L_1 , L_2 ; two capacitors C_1 , C_2 ; diode D ; H-bridge (4 switches S_1 – S_4).
- Model the shoot-through and non-shoot-through states.
- Define circuit parameters from the paper (inductance, capacitance, switching frequency, DC input voltage).

1.2 Model the PV parasitic capacitance (C_{PV}) and grid connection.

1.3 Define switching states and their corresponding CMV expressions.

Phase 2: Modulation Strategy Implementation

2.1 Implement conventional unipolar and bipolar PWM.

2.2 Implement the proposed CMV-reduced PWM scheme.

- Identify and eliminate high-CMV switching states.
- Generate gate signals for S_1 – S_4 according to the modified modulation logic.
- Incorporate shoot-through intervals for voltage boost.

2.3 Implement comparison schemes (e.g., H5 modulation, HERIC-type) if discussed.

Phase 3: Circuit Simulation

3.1 Build the circuit model in MATLAB/Simulink (Simscape Power Systems) or PLECS or LTspice.

3.2 Run time-domain simulations for each modulation strategy.

- Simulation time: several grid cycles (e.g., 0.2 s at 60 Hz).
- Record: output voltage/current, CMV waveform, leakage current through C_{PV} .

3.3 Compute metrics.

- CMV peak and RMS values.
- Leakage current peak and RMS.
- Output voltage THD (total harmonic distortion).
- Efficiency and power quality metrics.

Phase 4: Results Reproduction

- 4.1 Reproduce CMV waveform comparison plots.
- 4.2 Reproduce leakage current waveforms.
- 4.3 Reproduce output voltage/current THD comparison.
- 4.4 Compare tabulated numerical results against paper.

Datasets

No external datasets; all data is generated from circuit simulations with parameters specified in the paper.

Dataset / Source	Type	Location / Notes
Circuit parameters	Specified in paper	Table of component values (L , C , f_{sw} , V_{DC} , etc.)
Switching state tables	Specified in paper	CMV expressions for each state combination
Grid parameters	Standard	120 V/240 V, 60 Hz (or as specified)

Models, Tools, and Software

Simulation Platforms

Tool / Library	Version	Purpose / URL
MATLAB/Simulink	R2021b+	Circuit simulation with Simscape Electrical https://www.mathworks.com/
PLECS	latest	Alternative power electronics simulator https://www.plexim.com/plecs
LTspice	latest	Free SPICE simulator (alternative) https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html
Python + PySpice	latest	Open-source SPICE interface https://pyspice.fabrice-salvaire.fr/
NumPy / SciPy	≥ 1.21	Numerical analysis of waveforms, FFT for THD
Matplotlib	≥ 3.4	Waveform plotting

Hardware Requirements

- Standard workstation; circuit simulations are not computationally demanding.